

BHOPAL, IndiaWMO: **426670**Lat: **23.287N**Lon: **77.337E**Elev: **524**StdP: **95.19**Time Zone: **5.50 (IND)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.2	10.5	-1.9	3.4	24.3	0.0	4.0	23.4	6.4	23.0	5.5	21.6	1.7	50	0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	12.6	42.0	21.9	40.8	21.5	39.6	21.4	26.3	31.2	25.9	30.5	25.5	29.9	4.7	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	21.7	27.9	24.9	21.3	27.5	24.6	20.8	27.1	86.2	31.4	84.0	30.2	82.2	29.9	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.8	6.7	6.6	43.7	1.3	1.1	5.7	44.4	4.9	45.1	4.1	45.7	3.2	46.4		
(5)			4.4	27.9	1.3	1.1	3.4	28.7	2.6	29.4	1.9	30.0	0.9	30.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.7	17.9	21.1	26.1	31.0	33.8	31.3	26.9	25.9	26.8	25.9	22.6	19.0		
	DBStd	5.20	2.51	2.78	2.83	2.11	2.04	3.21	2.01	1.28	1.37	1.68	1.86	2.22		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	66	39	8	0	0	0	0	0	0	0	0	0	19		
	CDD10.0	5729	245	310	499	629	738	638	524	493	503	494	377	279		
	CDD18.3	2753	26	84	241	379	479	388	266	234	253	236	127	39		
	CDH23.3	33247	263	847	2936	5738	7836	5475	2431	1682	2237	2353	1082	368		
	CDH26.7	17714	39	295	1596	3671	5411	3309	783	350	797	1069	335	59		
(14) Wind	WSAvg	3.0	2.1	2.4	2.7	3.4	4.4	4.5	4.3	3.9	2.9	2.0	1.7	1.8		
(15) Precipitation	PrecAvg	1132	12	9	11	4	12	159	424	284	172	28	10	7		
	PrecMax	1905	66	49	89	28	67	461	783	877	524	126	111	114		
	PrecMin	796	3	0	0	0	0	24	96	33	30	0	0	0		
	PrecStd	251	15	15	20	6	19	109	140	180	117	33	24	23		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.4	33.5	38.5	41.9	43.6	43.2	35.7	32.2	34.5	35.3	32.3	29.8		
		MCWB	17.2	18.3	18.4	20.7	21.7	23.4	24.1	25.4	22.7	20.2	19.1	17.6		
	2%	DB	27.8	31.7	37.0	40.7	42.4	41.3	33.6	30.8	33.2	33.9	30.9	28.2		
		MCWB	16.6	17.4	18.3	20.1	21.8	23.1	24.4	25.2	23.4	20.8	18.8	17.0		
	5%	DB	26.3	30.0	35.6	39.7	41.4	39.5	32.1	29.7	32.0	32.8	29.8	26.9		
		MCWB	16.0	17.0	18.4	20.1	21.7	23.3	24.7	24.9	23.7	20.8	18.5	16.4		
	10%	DB	24.6	28.4	34.1	38.4	40.3	37.6	30.6	28.7	30.8	31.8	28.7	25.4		
		MCWB	15.2	16.6	18.1	19.8	21.5	23.5	24.7	24.8	24.0	20.4	18.2	15.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.3	20.3	22.3	25.2	27.0	26.7	27.0	26.6	26.4	25.4	23.2	20.0		
		MCDB	26.1	28.7	32.6	38.5	37.9	33.9	30.8	29.8	29.8	30.4	28.2	26.2		
	2%	WB	18.0	19.2	20.8	22.9	25.2	26.0	26.2	26.0	25.7	24.6	22.0	18.8		
		MCDB	24.6	27.8	32.1	36.9	36.7	33.1	29.8	29.1	29.3	29.4	26.3	25.3		
	5%	WB	17.1	18.4	19.8	21.7	24.2	25.5	25.8	25.5	25.2	23.8	20.7	17.7		
		MCDB	23.7	26.8	31.5	35.8	35.8	32.4	29.1	28.4	28.9	28.3	26.0	24.0		
	10%	WB	16.1	17.5	18.9	20.8	23.3	25.0	25.3	25.1	24.9	23.0	19.6	16.7		
		MCDB	22.6	26.0	31.1	35.3	35.2	31.8	28.4	27.6	28.4	27.5	25.9	23.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.6	13.4	14.4	14.6	12.6	9.8	5.6	4.9	7.6	11.9	12.9	13.0		
		MCDBR	14.6	15.2	15.9	15.6	13.3	11.9	8.5	6.9	10.1	13.2	13.7	14.6		
	5% WB	MCWBR	6.2	5.7	5.7	5.5	4.8	4.0	2.9	2.8	3.3	4.5	5.5	6.1		
		MCWBR	12.1	12.9	14.3	14.2	12.0	9.6	6.2	5.7	7.1	9.3	11.1	12.4		
(40) Clear-Sky Solar Irradiance	taub	taub	0.435	0.439	0.474	0.529	0.579	0.649	0.620	0.564	0.518	0.505	0.503	0.458		
		taud	2.063	2.089	2.009	1.912	1.843	1.751	1.826	1.982	2.077	2.009	1.941	2.014		
	Ebn at Noon	Ebn at Noon	813	843	831	791	747	692	712	754	783	770	737	769		
		Edh at Noon	150	156	176	198	211	230	213	182	163	167	168	152		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	4.33	5.26	6.30	7.01	7.08	5.61	3.81	3.72	4.89	5.29	4.54	4.21		
	RadStd	RadStd	0.31	0.24	0.27	0.24	0.17	0.60	0.58	0.47	0.53	0.29	0.27	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+2.42	N/A	+0.33	+0.44	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	+2.42	N/A	+0.33	+0.44	N/A	N/A	N/A	N/A

Nomenclature: See separate page